Lower Respiratory Diseases

NURA 340 Pathophysiology | Lecture Study Guide

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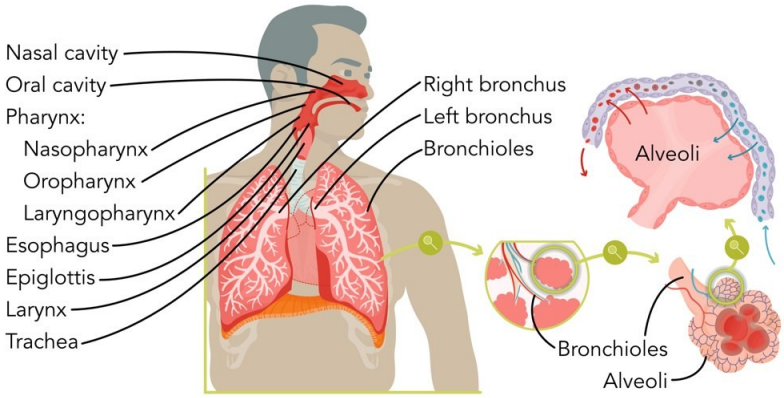
Lecture roadmap: Infectious agents and transmission → Acute bronchitis → Bacterial, viral, and fungal pneumonia → Tuberculosis → COPD → Traumatic lung injury (pneumothorax, tension pneumothorax, hemothorax) → Asthma

Building on the upper airway lectures, this lecture covers disorders of the trachea, bronchi, bronchioles, alveoli, and pleural spaces. Upper airway conditions were mostly self-limiting and often managed at home. Lower respiratory disorders affect respiratory muscle use, gas exchange, and fluid balance, so exacerbations usually mean a visit to a provider, possible hospitalization, and sometimes severe or critical illness.

Core pillars to track in every condition: inflammation, immunity and infection, fluid and electrolyte balance, and acid-base balance. Trace each condition from tissue injury → manifestation → gas exchange failure → systemic acid-base or fluid shift.

Introduction to Infectious Respiratory Diseases

Topic	Key points
Viruses	Influenza, RSV, seasonal coronaviruses, COVID-19, SARS
Bacteria	Streptococcus pneumoniae, Haemophilus influenzae, Mycoplasma pneumoniae (very common)
Fungi	Fungal pathogens (typically opportunistic, see fungal pneumonia)
Droplet transmission	Large respiratory secretions from coughing or sneezing that settle quickly
Airborne transmission	Microscopic particles that stay suspended for long periods and are inhaled deep into the distal airways
Direct or indirect contact	Direct physical transmission, or touching a contaminated surface followed by self-inoculation

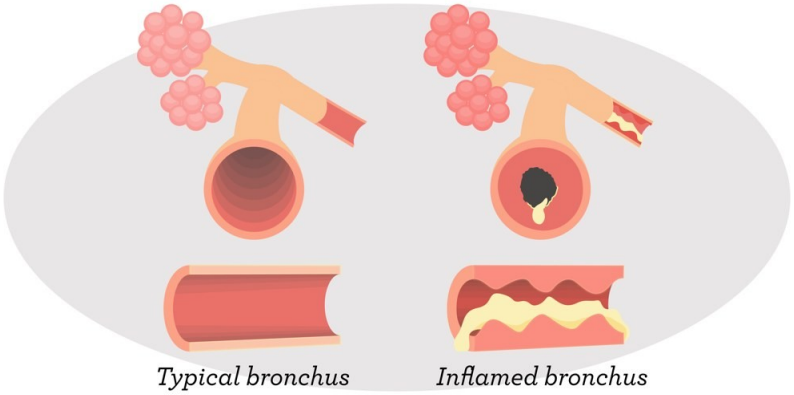


Respiratory tract: where lower airway infections occur (lecture slide)

Awareness, not tested: Know which patients may be in isolation and what type. Droplet precautions: gown, gloves, mask, and a face shield to protect the eyes. Airborne precautions: an N95 respirator instead of a standard mask. Contact transmission is covered by the same PPE used in the respiratory isolation precautions, so no additional PPE is needed.

Acute Bronchitis

Typically self-limiting inflammation of the large tracheobronchial airways.



Typical bronchus vs. inflamed bronchus (lecture slide)

Etiology

- **Viral:** over 90% of cases (rhinovirus, adenovirus, influenza, parainfluenza). Apart from influenza, these are the "common cold" viruses.
- **Bacterial:** rare, and usually seen only with underlying lung disease from smoking or long-term inhaled chemicals.

Pathophysiology

1. **Infection:** pathogens infect the bronchial epithelial cells when inhaled and trigger an acute inflammatory response.
2. **Vascular changes:** inflammatory cytokines cause microvascular dilation and submucosal edema.
3. **Mucus:** hypertrophied goblet cells hypersecrete mucus into the bronchial lumen.

4. **Impaired clearance:** epithelial injury causes ciliary dyskinesia and destruction, which impairs the mucociliary escalator.
5. **Result:** mucus accumulation, swelling, edema, and narrowing of the bronchial lumen.

Manifestations

Manifestation	Mechanism
Productive cough	Mucus and cellular debris stimulate bronchial vagal sensory cough receptors. A productive cough clears the blocked airways
Wheezing	Air forced through narrow, edematous bronchi: high-pitched musical sound
Rhonchi	Turbulent air against thick secretions in large airways: low-pitched, continuous rumbling
Dyspnea	Narrowed lumens increase airway resistance and the work of breathing. Retractions are usually absent because the extra work is mild
Low-grade fever, fatigue, malaise, muscle aches	Circulating inflammatory cytokines alter hypothalamic thermoregulation and drive the systemic inflammatory response

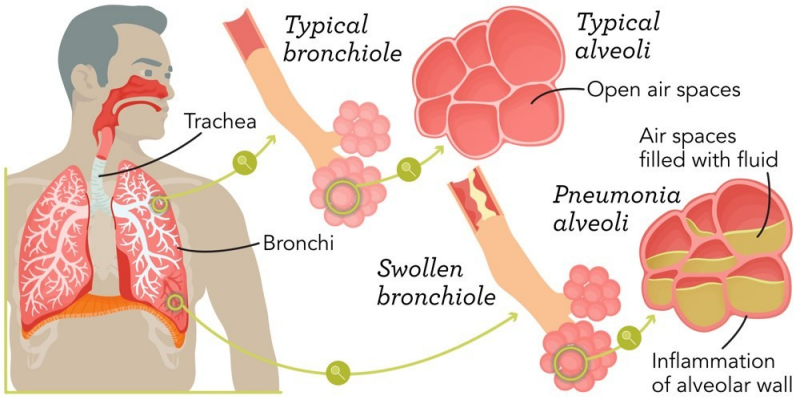
Exam note: You may not be able to tell the lung sounds apart, but expect to hear wheezing and crackles in your patient.

Diagnosis and Course

- **Clinical diagnosis** from history and physical exam. The goal is to rule out pneumonia by listening to lung sounds. If needed, a chest X-ray will be negative for pneumonia.
- **No further workup** if vital signs are in expected ranges and the exam does not suggest pneumonia. Patients go home with anti-inflammatory medication and rest.
- **Resolves** in about 2 to 3 weeks without lasting effects. The cough can linger 10 to 20 days after the inflammation is gone, and the patient is no longer contagious.

Pneumonia

Very common, especially in hospitalized patients. The three types differ in where the inflammation sits and how it presents.



Bronchioles and alveoli in pneumonia: fluid-filled air spaces (lecture slide)

Bacterial Pneumonia

- **Common causes:** Streptococcus pneumoniae (most common), Staphylococcus aureus, Klebsiella pneumoniae, Pseudomonas aeruginosa. Focus on the first three.
- **Risk factors:** people with chronic conditions or lifestyle risks rather than healthy lungs: COPD and other cardiopulmonary conditions, smoking, alcohol use disorder, poor dental hygiene, immunosuppression, aspiration risk, and asthma. Aspiration means food or liquid goes into the lungs and pools there, letting bacteria grow even in small amounts.

Pathophysiology

1. **Colonization:** inhaled or aspirated bacteria bypass the upper airway defenses and colonize the alveoli, where they proliferate and spread.
2. **Innate response:** alveolar macrophages release pro-inflammatory cytokines, causing capillary vasodilation and increased endothelial permeability.
3. **Consolidation:** protein-rich exudate and neutrophils flood the alveolar spaces. This is the "consolidation" seen on X-rays.
4. **Narrowing:** alveolar walls become inflamed and swollen, and surrounding bronchioles narrow.

Manifestations

Manifestation	Mechanism
Fever and chills	Bacterial endotoxins and endogenous pyrogens reset the hypothalamic thermostat. Involuntary shivering (chills) raises core temperature, and the fever helps kill bacteria that cannot survive heat. Fever can be high
Productive cough, purulent sputum	Dying neutrophils, bacteria, and fibrin in the alveolar exudate trigger cough. Sputum is green, yellow, or rust-colored. Rust color comes from red blood cells leaking through inflamed air sacs
Dyspnea and hypoxia	Fluid-filled alveoli cannot exchange gas. This creates an intrapulmonary shunt: perfused alveoli without ventilation. Low oxygen drives dyspnea, increased work of breathing, and mental status changes
Early respiratory alkalosis	Hypoxemia triggers compensatory tachypnea that blows off CO2
Later respiratory acidosis	When the patient tires, or fluid and sputum block the bronchioles and trap CO2
Fluid loss	High fever and tachypnea increase insensible losses, and exudate pulls fluid out of the vasculature. Watch hydration

Acid-base pattern: Pneumonia often starts as respiratory alkalosis (tachypnea) and can transition to respiratory acidosis as the patient fatigues. The ABG shows where the patient is on that path.

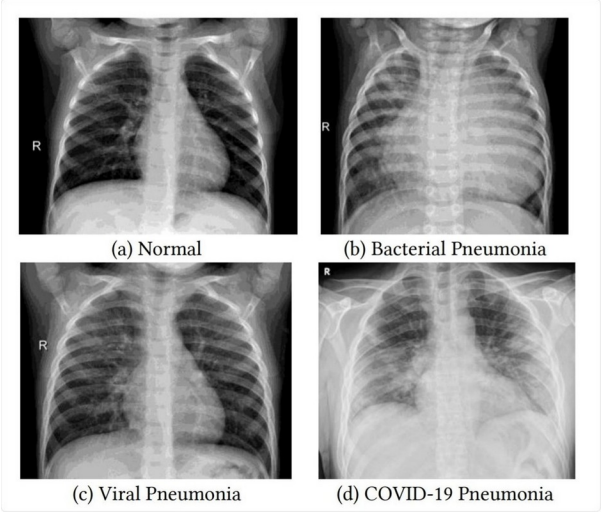
Diagnosis and Complications

- **Chest X-ray:** infiltrates or consolidations. Bacterial, viral, and fungal pneumonias look different on X-ray.
- **Sputum culture** is positive and identifies the organism.
- **Elevated WBC count** shows the inflammatory response. Blood gas analysis shows oxygenation and acid-base status.
- **Complications:** respiratory failure requiring mechanical ventilation, lung abscess, and sepsis with multi-organ dysfunction. Recognize and treat quickly.
- **Older adults** are at higher risk for complications because of age, functional status, pulmonary disease, immunosuppression, and comorbidities.

Viral Pneumonia

- **Common causes:** influenza A and B, RSV, adenovirus, COVID-19. Upper respiratory viruses can breach the defenses and reach the lower lungs.
- **Risk factors:** infants, older adults, pregnancy, immunosuppression, and chronic medical conditions.

Unlike bacterial pneumonia, which fills the alveolar sacs with purulent exudate, viruses invade and replicate in the ciliated columnar epithelial cells of the airway and the pneumocytes of the alveolar walls. This causes interstitial inflammation: edema and mononuclear cell infiltration in the alveolar septum and interstitial walls, rather than dense consolidation inside the alveoli.



Chest X-rays: normal, bacterial, viral, and COVID-19 pneumonia (lecture slide)

Onset: Viral pneumonia usually has a gradual onset with a slower build of manifestations.

Manifestation	Mechanism
Fever and fatigue	Systemic immune response
Dry, non-productive cough	Inflammatory fluid is in the alveolar walls, not the air sacs, so there is minimal exudate to mobilize. The cough persists but brings nothing up
Chest pain with coughing	Interstitial stretching and thoracic muscle strain from repetitive coughing irritate chest wall nerve endings

- **Diagnosis:** chest X-ray (bilateral infiltrates showing widespread fluid and inflammation, versus consolidation on one side in bacterial), viral PCR swab, no purulent drainage, WBC often in the normal range, possibly elevated inflammatory markers, and a negative bacterial culture. Bacterial testing is usually done at the same time to separate viral from bacterial.
- **Nurse scope:** nurses do not have to read X-rays, but should understand what the study can show and learn to recognize when a patient looks better or worse.
- **Complication:** acute respiratory distress syndrome (ARDS), particularly with influenza and severely ill COVID-19 patients. ARDS needs ICU admission and mechanical ventilation. Details come in a later course. Recognize patients who are not improving so care can be escalated.

Fungal Pneumonia

Typically an opportunistic infection seen in immunocompromised clients, from opportunistic or endemic fungal spores. The most common is Pneumocystis jirovecii pneumonia.

1. **Inhalation:** spores lodge in the distal alveoli.
2. **Germination:** in immunocompromised hosts, impaired cell-mediated immunity lets spores germinate and invade the lung parenchyma and pulmonary microvasculature.
3. **Tissue damage:** invasion causes granulomatous inflammation, tissue necrosis, and microvascular erosion.

Severity: Manifestations are more severe and progress faster because the immune response is impaired.

Manifestation	Mechanism
Dyspnea, tachypnea, tachycardia	Interstitial swelling and granuloma formation decrease lung compliance, triggering compensatory tachypnea and reflex tachycardia
Rales (crackles)	Fluid in the small airways makes them pop open during inspiration. Rales are crackles, and crackles are common in pneumonia, especially bacterial
Pleural rub	Fungal extension to the visceral pleura causes fibrous inflammation, and the inflamed pleural surfaces rubbing together make the sound
Hemoptysis	Fungi erode pulmonary blood vessels, and blood leaks into the airways. Usually red-tinged or dark red or rust-colored secretions rather than large amounts

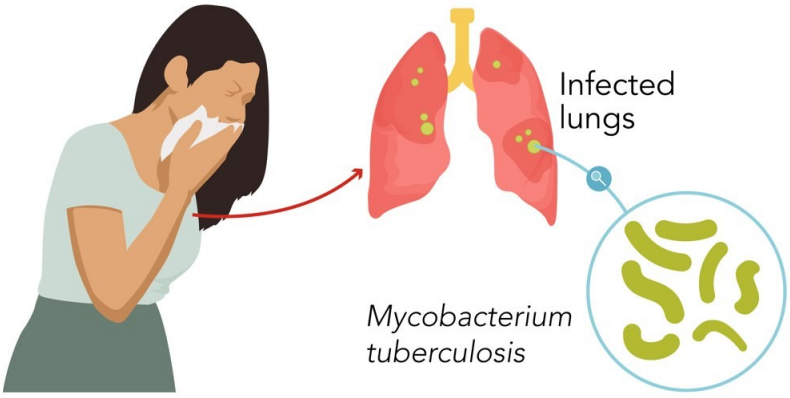
- **Diagnosis:** epidemiological exposure, host risk factors (immunocompromise), chest X-ray patterns, and fungal cultures. In a non-immunocompromised patient, fungal testing is considered when pneumonia does not resolve and viral and bacterial causes have been ruled out. Biopsy or special CT (upper lobe nodules) may be used.
- **Complications:** blindness, cerebral emboli or infarction, myocardial or pulmonary infarctions, and dissemination to other sites. Spread is easier in immunocompromised patients, so treat early.

Pneumonia Comparison

	Bacterial	Viral	Fungal
Where	Inside alveoli (consolidation)	Alveolar walls and interstitium	Parenchyma, granulomas, pleura, vessels
Onset	Often abrupt, high fever and chills	Gradual	Faster and more severe if immunocompromised
Cough	Productive, purulent (green, yellow, rust)	Dry or minimally productive	Nonproductive, possible hemoptysis
X-ray	Consolidation, often one side	Bilateral infiltrates	Varies; upper lobe nodules on CT
WBC	Elevated	Often normal	Depends on host
Culture	Sputum culture positive	Bacterial culture negative; viral PCR	Fungal culture
Who	COPD, smokers, aspiration, immunosuppressed	Infants, older adults, pregnancy	Immunocompromised

Tuberculosis (TB)

A chronic granulomatous infection caused by Mycobacterium tuberculosis, transmitted by airborne droplets. In the hospital this requires an N95 respirator. The bacterium is the cause, so the name is the easy part to remember.

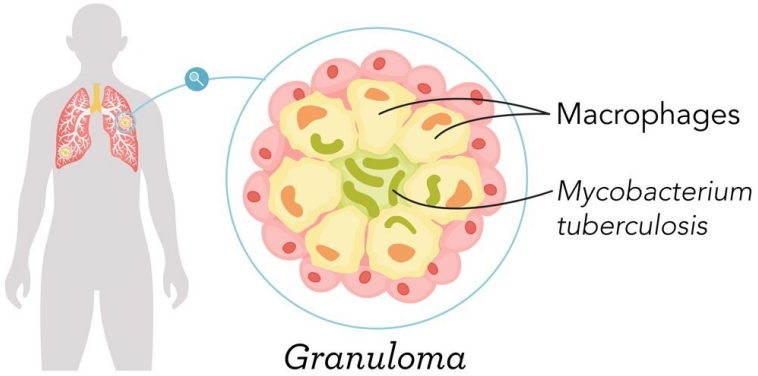


Transmission of Mycobacterium tuberculosis and infected lungs (lecture slide)

- **Risk factors:** immunocompromise, diabetes, cancer, renal failure, transplant patients, children under 5, and substance use disorders.
- **Social determinants:** medically underserved populations and crowded living conditions have higher rates, not because the people are inherently at higher risk but because of limited treatment, poorly maintained housing, hygiene practices, and difficulty isolating. Because spread is airborne, if one person gets TB, others in the space can too.

Pathophysiology

1. **Engulfment:** the inhaled bacillus reaches the alveoli and is engulfed by alveolar macrophages.
2. **Evasion:** its lipid-rich cell wall prevents lysosomal fusion, so the bacteria replicate inside the macrophage.
3. **Granuloma:** cell-mediated immunity (T cells) recruits macrophages to surround the infected cells, forming a granuloma with a central caseous necrosis.
4. **Latent TB:** the granuloma wall keeps the bacteria contained and not replicating.
5. **Active TB:** immunosuppression breaks down the granuloma wall, releasing bacteria and causing cavitary destruction of lung tissue.



Granuloma: macrophages surrounding Mycobacterium tuberculosis (lecture slide)

Manifestations: Latent vs. Active

	Findings and mechanism
Latent TB	Asymptomatic. Bacteria are sequestered inside intact granulomas, so there is nothing to spread or damage. The patient still tests positive, and the X-ray shows latent disease
Active: low-grade fever, night sweats	Macrophages continuously release tumor necrosis factor and interleukins, causing temperature spikes and nocturnal diaphoresis
Active: unintentional weight loss	High circulating TNF accelerates protein breakdown and suppresses appetite
Extrapulmonary TB	Bacteria spread through the blood and lymphatics to lymph nodes, kidneys, bones, and meninges. Manifestations vary with the site

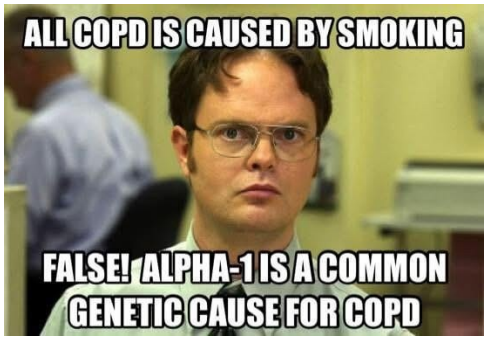
Diagnosis

- **Skin test (PPD):** injected, then read in about 2 days. Blood test (IGRA): a single blood draw.
- **Positive skin test:** additional testing determines latent vs. active, or whether the patient just reacted to the skin test itself.
- **Sputum smear or culture** helps determine latent vs. active. Chest X-ray identifies fluid and lesions.
- **Outcomes:** active TB brings a gradual but significant decline in function from persistent respiratory symptoms, chronic fatigue, and wasting (appetite suppression). Permanent lung damage is possible.

Chronic Obstructive Pulmonary Disease (COPD)

Very common in adults, and one to know well because many adult patients eventually have it. COPD is not a single disease but a group of progressive, irreversible lung diseases that includes chronic bronchitis and emphysema. The two often coexist.

- **Cause:** long-term exposure to noxious gases and particles: tobacco smoke (most common), biomass fuel smoke, and chemical exposure at work.
- **Genetic:** alpha-1 antitrypsin deficiency also increases risk.



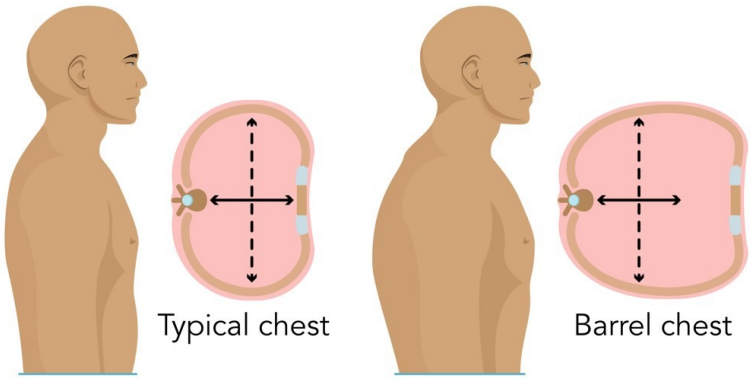
Not all COPD is caused by smoking: alpha-1 antitrypsin deficiency (lecture slide)

Phenotype	Mechanism
Chronic bronchitis	Chronic inhalation of irritants causes goblet cell hyperplasia, mucus hypersecretion, and airway wall fibrosis. This leads to persistent airflow obstruction, and air is harder to get out than in
Emphysema	Chronic inflammation recruits proteases such as elastase, which destroy alveolar walls and elastic recoil fibers. The distal airways collapse during exhalation, trapping air in the alveoli and causing hyperinflation

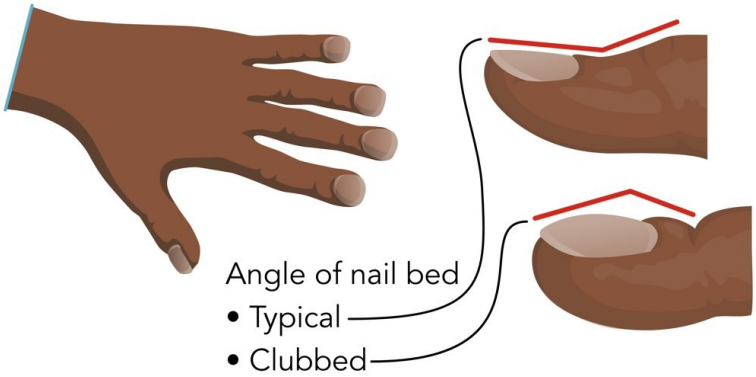
COPD develops gradually, not as a sudden new onset. Some signs are seen at baseline and others during exacerbations. Exacerbations can come from exercise, more irritant exposure, or any upper respiratory infection. Patients usually come to the hospital because of an exacerbation, not just because they have COPD, since it can be managed at home by avoiding irritants.

Manifestations

Manifestation	Mechanism
Dyspnea, chronic cough, sputum	Develop over time as irritant exposure continues
Accessory muscle use, tachypnea	Dynamic hyperinflation flattens the diaphragm, making it mechanically inefficient. The patient recruits neck (sternocleidomastoid) and chest muscles and raises the rate
Decreased breath sounds	Alveolar destruction and hyperinflated air pockets muffle airflow
Chronic hypoxia	Alveolar wall destruction reduces the surface area for gas diffusion
Chronic hypercapnia	Airway collapse prevents clearance of expired air, so CO_2 is retained
Fatigue and weight loss	High work of breathing uses much of the daily energy budget, with higher metabolic demand and more calories burned just to breathe
Barrel chest	Chronic air trapping and hyperinflation permanently enlarge the anterior-posterior diameter and alter rib cage geometry
Clubbing	Persistent peripheral hypoxemia stimulates local platelet-derived growth factors, causing hypervascularity and connective tissue growth in the nail bed and changing the nail angle



Typical chest vs. barrel chest (lecture slide)



Typical vs. clubbed nail bed angle (lecture slide)

Key concepts: (1) COPD patients live at a lower baseline oxygen level, often below 95%. During an exacerbation on oxygen, the target SpO2 is 88% to 92%. (2) Chronic CO2 retention causes compensated respiratory acidosis: over days the kidneys conserve bicarbonate to keep pH near normal. (3) Barrel chest and clubbing do not show the severity of an exacerbation. They show long-standing chronic hypoxemia and hyperinflation.

Diagnosis and Impact

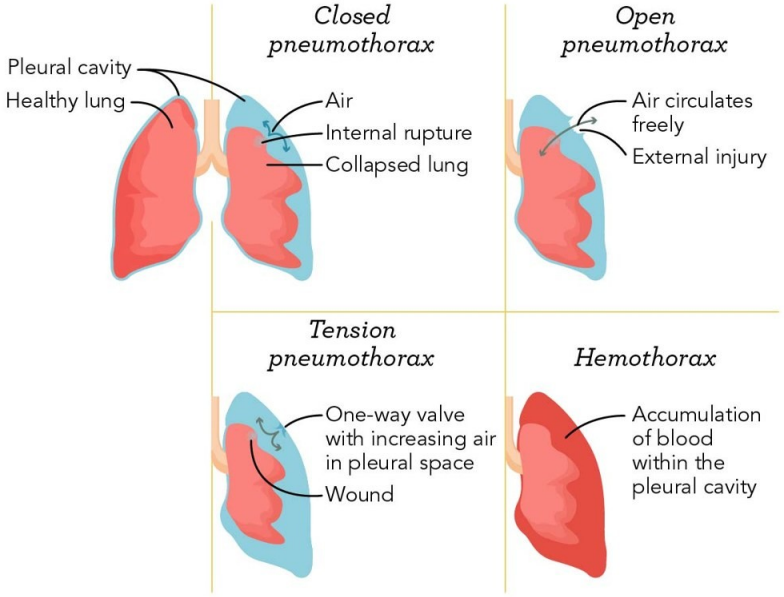
- **Pulmonary function tests:** essential for diagnosis, staging, and long-term assessment.
- **Chest X-ray or CT:** anatomical changes, lung capacity, and related heart abnormalities.
- **Labs:** CBC for infection or anemia. ABG for exacerbation acuity: how severe the respiratory acidosis is, and whether the kidneys have been able to compensate yet.
- **Quality of life:** limited activity, sleep disrupted by recurrent productive cough (chronic bronchitis), low energy, higher risk of respiratory infection, and a higher risk of anxiety, depression, and social isolation.
- **Complications:** pneumonia (common in exacerbations), pulmonary hypertension, and eventually heart failure.

Traumatic Lung Injury and the Pleural Space

Traumatic lung injury is a disruption of the pleural membrane and underlying lung parenchyma from blunt or penetrating trauma, or from internal bleb rupture. Normally the pleural space holds a negative

intrapleural pressure that keeps the elastic lung expanded against the chest wall. Injury can cause respiratory distress, low oxygen, and cardiovascular compromise if not recognized and treated.

Memory tip: The types all end in "thorax": pneumothorax (closed, open, tension) and hemothorax.

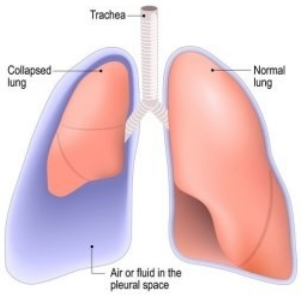


Closed, open, and tension pneumothorax, and hemothorax (lecture slide)

Type	Definition
Closed pneumothorax	Air enters the pleural space through an internal lung rupture with no external chest wound
Open pneumothorax ("sucking chest wound")	Penetrating chest trauma breaks the parietal pleura and lets atmospheric air move freely in and out of the pleural cavity, equalizing intrapleural and atmospheric pressure
Tension pneumothorax	Most critical. A flap injury acts as a one-way valve: air enters the pleural space on inspiration but cannot escape on exhalation, so intrapleural pressure skyrockets
Hemothorax	Accumulation of blood in the pleural cavity, usually from laceration of intercostal or internal mammary vessels

Spontaneous Pneumothorax

A rupture of a subpleural apical air bleb. This is a closed pneumothorax with no clear cause, common in tall, thin, young adolescent and adult males (for example cross-country runners and soccer players who suddenly have chest pain and harder breathing). It can also occur with underlying lung disease such as COPD.



Collapsed lung from air in the pleural space (lecture slide)

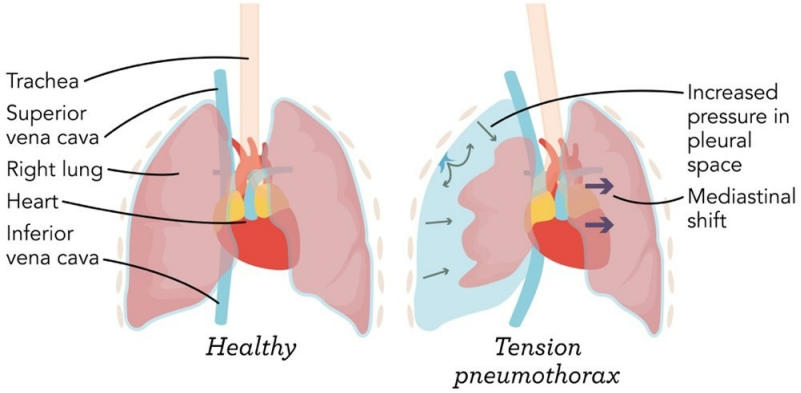
A bleb ruptures and breaches the visceral pleura, letting lung air enter the intrapleural space. As intrapleural pressure rises toward atmospheric, the negative pressure gradient holding the lung open is lost, and the lung collapses inward from its natural elastic recoil.

Manifestation	Mechanism
Sudden pleuritic chest pain	Air stretches the sensory nerve fibers of the parietal pleura, causing sharp pain with breathing
Dyspnea, tachycardia	Reduced vital capacity and functional lung volume cause hypoxemia, which triggers sympathetic outflow and raises heart and respiratory rate
Decreased breath sounds and tactile fremitus	Air in the pleural space acts as an acoustic insulator, dampening vocal vibrations and breath sounds over the affected lung
Hyperresonance	Percussion over the air-filled pleural space gives a booming or hollow sound

- **Diagnosis:** upright chest X-ray showing the visceral pleural line and absence of lung markings. Point-of-care ultrasound can be used in severe cases in the ER or trauma setting.
- **Outcomes:** primary (healthy individuals) has about a 30% risk of recurrence within 5 years, respiratory compromise, and possible prolonged air leak. Secondary (such as COPD patients) can bring chronic pain, reduced lung function, and anxiety about recurrence.

Tension Pneumothorax

Priority: A tension pneumothorax is a life-threatening medical emergency and always the priority patient. It is an airway issue (ABCs). It can rapidly lead to obstructive shock, with tachycardia, hypotension, jugular vein distension, cardiopulmonary collapse, and death unless the air is immediately decompressed.



Healthy chest vs. tension pneumothorax with mediastinal shift (lecture slide)

- **Causes:** penetrating or blunt chest trauma, mechanical ventilation barotrauma (lung injury from ventilator pressure), and an open pneumothorax left untreated.

1. **One-way valve:** a tissue flap lets air in on inhalation and seals shut on exhalation.
2. **Pressure builds:** progressive air trapping creates massive positive intrapleural pressure that collapses the ipsilateral lung completely.
3. **Mediastinal shift:** the heart, trachea, and major vessels are pushed toward the unaffected side.
4. **Venous compression:** high mediastinal pressure compresses the thin-walled superior and inferior vena cava, blocking venous return to the right atrium and causing hemodynamic collapse.

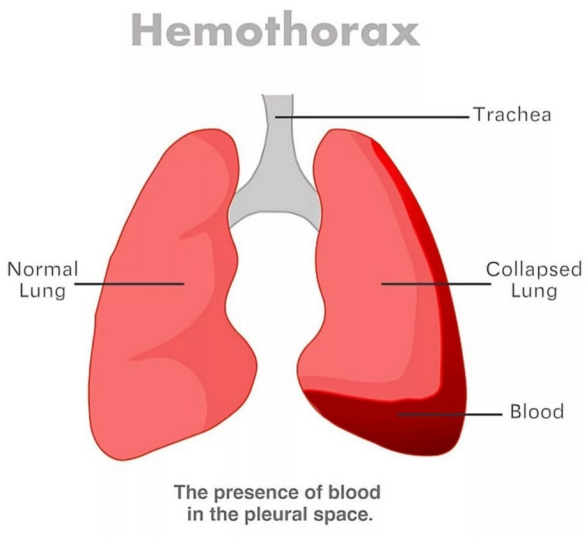
Manifestation	Mechanism
Severe respiratory distress	Complete collapse of the affected lung, plus compression of the healthy lung as chest structures are pushed into it
Hypotension	Compressed vena cava lowers preload, which lowers stroke volume and cardiac output
Jugular vein distension	Venous blood backs up behind the compressed vena cava, raising central venous pressure and distending the neck veins
Tracheal deviation (hallmark)	Pressure physically pushes the flexible trachea toward the opposite, uninjured side along with the heart and vessels

Exam note: Tracheal deviation is the hallmark sign of tension pneumothorax.

- **Diagnosis:** clinical. There is no time to wait for a chest X-ray, or the patient will code. Severe distress, visible tracheal deviation, hypotension, and absent breath sounds on one side lead to immediate needle decompression. You do not need to know the procedure for this class, only that treatment starts before an X-ray.
- **Complications:** respiratory failure or arrest, cardiac arrest, and death if untreated.

Hemothorax

Blood in the pleural space, usually from penetrating or blunt chest trauma, intercostal vessel laceration, or complications of thoracic surgery. Chest injury with bleeding into the chest cavity is the most common cause.



Hemothorax: blood in the pleural space (lecture slide)

Blood occupies the space normally reserved for lung expansion. Gravity pools it in the lower parts of the cavity. It compresses lung tissue, causing mechanical atelectasis, while removing blood volume from circulation. On X-ray a hemothorax appears at the base working upward, whereas air (lighter than liquid) in a pneumothorax can sit higher in the pleural space.

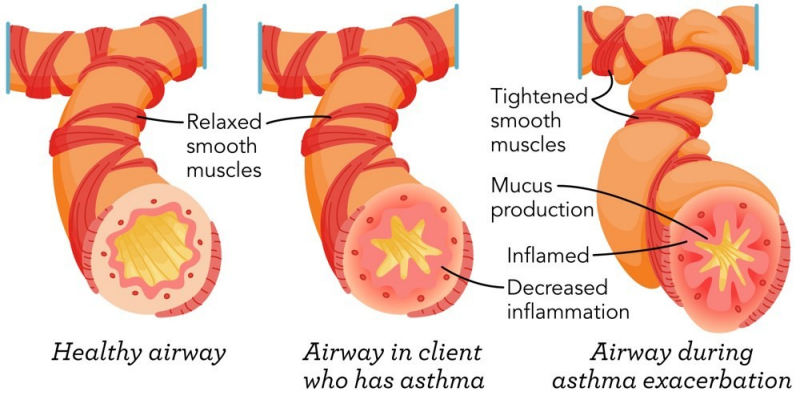
Manifestation	Mechanism
Dyspnea, diminished breath sounds on the affected side	Same respiratory findings as pneumothorax
Dullness to percussion	Fluid density
Hypotension, tachycardia	Reduced intravascular volume (blood now in the pleural space) can progress to hypovolemic shock. Tachycardia maintains tissue perfusion
Cool, pale, clammy skin	Compensatory sympathetic surge causes peripheral vasoconstriction to raise blood pressure

- **Diagnosis:** chest X-ray shows pleural effusion and a fluid level. It cannot tell hemothorax from a pleural effusion (fluid without necessarily a collapsed lung). It is confirmed only when drained by chest tube or thoracentesis and frank blood comes out. Hypotension, tachycardia, and cool clammy skin make it highly suspected.
- **Fluid balance:** acute intrapleural hemorrhage causes intravascular volume depletion and isotonic fluid loss (hypovolemia).
- **Concerns:** respiratory compromise and hemodynamic instability. Untreated, continuing blood accumulation with no exit can cause mediastinal shift and other serious complications.

Asthma

Asthma was covered in the immunity and inflammation lectures but is repeated here because it is tested with the respiratory content. Many people think of it as a respiratory disorder and group it that way, but ATI does not.

Asthma is a chronic inflammation of the airway characterized by reversible hyperresponsiveness. It is influenced by genetic and environmental factors, and exacerbations are triggered by allergens, viral infections, exercise, and cold air.



Healthy airway vs. airway in asthma vs. asthma exacerbation (lecture slide)

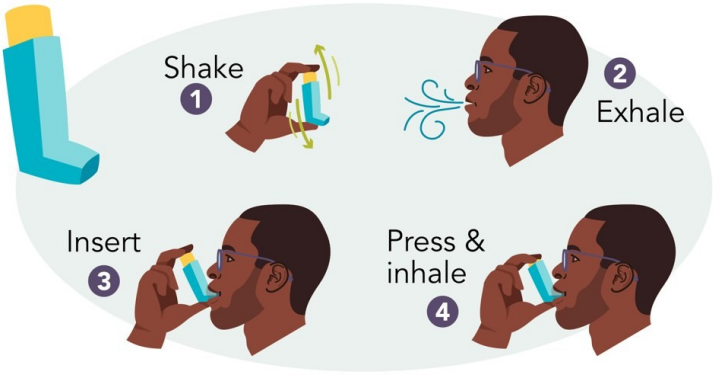
Pathophysiology

1. **Trigger:** exposure to an allergen or irritant starts a type I hypersensitivity response.
2. **Degranulation:** the allergen cross-links on mast cells, causing immediate degranulation and release of histamine, leukotrienes, and prostaglandins.
3. **Early phase (first minutes):** histamine and leukotrienes cause immediate airway smooth muscle contraction (bronchospasm).
4. **Late phase (after a few hours):** cytokines recruit eosinophils and T helper cells, leading to submucosal edema and mucus hypersecretion.
5. **Result:** plugged and constricted airways during the exacerbation.

Manifestations

Manifestation	Mechanism
Wheezing (hallmark)	Expiratory air forced through bronchospastic, edematous, mucus-plugged airways creates turbulent musical sounds
Chest tightness	Inflamed airways make it hard to get air in. The patient reports the tightness; the nurse cannot feel it in the muscles
Cough	Mucus plugs and local inflammatory mediators stimulate sensory receptors
Severe: confusion, extreme tiredness	Signal status asthmaticus and impending respiratory muscle fatigue, with progressive hypoventilation, severe hypoxemia, and hypercapnic encephalopathy. The confusion and lethargy are often from high CO2, not just low oxygen

Status asthmaticus: The exacerbation does not respond to the patient's home medications or to additional ER medications, and the patient must stay in the hospital. It can be deadly: inflamed, mucus-filled, edematous airways trap air that is very hard to exhale, and CO2 rises. Treat before it gets to that point.



Using an inhaler: shake, exhale, insert, press and inhale (lecture slide)

Diagnosis

- **Pulmonary function tests with spirometry showing reversibility:** the test is repeated after a short-acting beta agonist such as albuterol. Lung function improves, either back to normal or better than before the medication.
- **Age:** asthma is not diagnosed until a child is at least 2 years old. Under 2, it is called reactive airway disease. It is treated the same.

Acid-base: Early acute asthma: hyperventilation lowers CO2, giving respiratory alkalosis. Late and severe asthma: CO2 rises. A normal CO2 may be a swing back, and an elevated CO2 is an alarm sign that the muscles are fatiguing and a life-threatening respiratory acidosis is starting.

Impact and Associated Conditions

- **Burden:** medical costs and missed work. Children may miss school, affecting social development and academic progress.
- **Associated conditions:** pneumonia (from an exacerbation), allergic rhinitis, GERD, obstructive sleep apnea, and anxiety. Asthma patients are at higher risk for other chronic disorders, especially with a genetic component.

Quick Review

Condition	Key points
Acute bronchitis	>90% viral; inflammation of large airways with mucus; productive cough, wheezes/rhonchi; clinical diagnosis, X-ray negative for pneumonia; resolves in 2 to 3 weeks, cough up to 10 to 20 days
Bacterial pneumonia	S. pneumoniae most common; alveolar consolidation; fever, chills, purulent green/yellow/rust sputum; shunt and hypoxemia; early resp. alkalosis then acidosis; sputum culture, high WBC
Viral pneumonia	Influenza, RSV, COVID-19; interstitial inflammation; gradual onset, dry cough, chest pain with cough; bilateral infiltrates, normal WBC; can progress to ARDS
Fungal pneumonia	Immunocompromised (Pneumocystis jirovecii); granulomas, necrosis, vessel erosion; rales, pleural rub, hemoptysis; fungal culture; dissemination
Tuberculosis	M. tuberculosis, airborne (N95); granuloma with caseous necrosis; latent = asymptomatic and contained; active = fever, night sweats, weight loss; PPD or blood test, sputum, CXR
COPD	Chronic bronchitis + emphysema; irreversible; air trapping; SpO2 target 88 to 92%; chronic hypoxia/hypercapnia with compensated respiratory acidosis; barrel chest, clubbing; PFTs and ABG
Spontaneous pneumothorax	Apical bleb rupture, tall thin young males or COPD; sudden pleuritic pain, decreased breath sounds, hyperresonance; upright CXR; 30% recurrence in 5 years
Tension pneumothorax	Emergency; one-way valve; tracheal deviation (hallmark), JVD, hypotension, severe distress; clinical diagnosis, treat before X-ray
Hemothorax	Blood in pleural space; pools at base; dullness to percussion, hypotension, tachycardia, cool clammy skin; confirmed when drained (frank blood)
Asthma	Type I hypersensitivity; reversible hyperresponsiveness; wheezing (hallmark), chest tightness, cough; status asthmaticus = confusion/fatigue; spirometry reversibility; early alkalosis, late acidosis (rising CO2 is an alarm sign)

Notes on Transcript Cleanup

Corrected mis-transcriptions: "Streptococcus pneumonia" is Streptococcus pneumoniae; "Mycoplasma pneumonia" is Mycoplasma pneumoniae; "cautious necrosis" is caseous necrosis; "infractions" is infarctions; "mucilary" is mucociliary; "hemothorax, sorry" in the tension pneumothorax section is a spoken slip for pneumothorax and was written as pneumothorax. "Fast-acting" Mycobacterium tuberculosis is kept as spoken (the organism is actually slow-growing). The PPD was described as a subcutaneous injection; it is given intradermally. A garbled phrase at "cue six" in the COPD breath

sounds explanation was removed. The asthma diagnosis sentence about the pulmonary function test before and after albuterol was garbled in the transcript and was written to match the slide meaning (lung function improves after a short-acting beta agonist). The explanation of rhonchi vs. crackles was inconsistent in the lecture, so the instructor's practical advice is kept: expect wheezing and crackles, and rales are crackles.